

Quiz 8

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find the solution of the following initial value problem:

$$y'' + 8y' + 25y = 0, \quad y(0) = 1, \quad y'(0) = -1.$$

Characteristic polynomial:

$$\begin{aligned} r^2 + 8r + 25 &= 0 & y(x) &= e^{-4x}(C_1 \cos 3x + C_2 \sin 3x) \\ \Rightarrow r &= \frac{-8 \pm \sqrt{8^2 - 100}}{2} & \Rightarrow \text{den} & \\ &= -4 \pm 3i & y'(x) &= e^{-4x}(-3C_1 \sin 3x + 3C_2 \cos 3x) \\ & & & -4e^{-4x}(C_1 \cos 3x + C_2 \sin 3x) \end{aligned}$$

$$\begin{aligned} y(0) &= 1 \Rightarrow C_1 = 1 \\ y'(0) &= -1 \Rightarrow 3C_2 - 4C_1 = -1 \end{aligned} \quad \left\{ \begin{aligned} C_1 &= 1 \\ C_2 &= 1 \end{aligned} \right.$$

Thus $y(x) = e^{-4x}(\cos 3x + \sin 3x)$

Problem 2. (5 points) Find the general solution of the homogeneous differential equations

$$9y^{(3)} + 12y'' + 4y' = 0.$$

Characteristic polynomial:

$$\begin{aligned} 9r^3 + 12r^2 + 4r &= 0 \\ r(9r^2 + 12r + 4) &= 0 \\ r(3r + 2)^2 &= 0 \\ \Rightarrow r_1 &= 0, \quad r_2 = r_3 = -\frac{2}{3} \end{aligned}$$

Thus $y(x) = C_1 + e^{-\frac{2}{3}x}(C_2 + C_3x)$